pfi

Python for Fixed Income

COURSE 6 · NEURAL NETWORKS

Case Study: Sequence Models on the Treasury Curve

WEEK 3 · DAY 4

Created by **Jeff Lenamon**

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Choose first, then open once

1. Finish netkit

fold_comparison and assert_history

2. Choose first

an LSTM and a GRU; a rule written first

3. Open once

the 939 held-out dates, scored once

Nothing is chosen after the held-out years are read: the choice stands.

A walk-forward error on past dates describes those dates. It is not evidence about any later date.

How a fold comparison works

1. Same folds

a model and a baseline

2. Three reads

two means, their difference

3. A count

folds where the model is lower

4. The result

one short Series

You control

Which columns are the model and the baseline.

What comes out

Means, difference, folds below, folds.

Fold by fold: fold_comparison

```
rules = netkit.fold_comparison(baselines, model="train_mean_mae_bp", baseline="zero_change_mae_bp")
# model_mean, baseline_mean, difference (model - baseline), folds_below, folds
```

3.22 bp

training mean, mean of 5 folds

3.21 bp

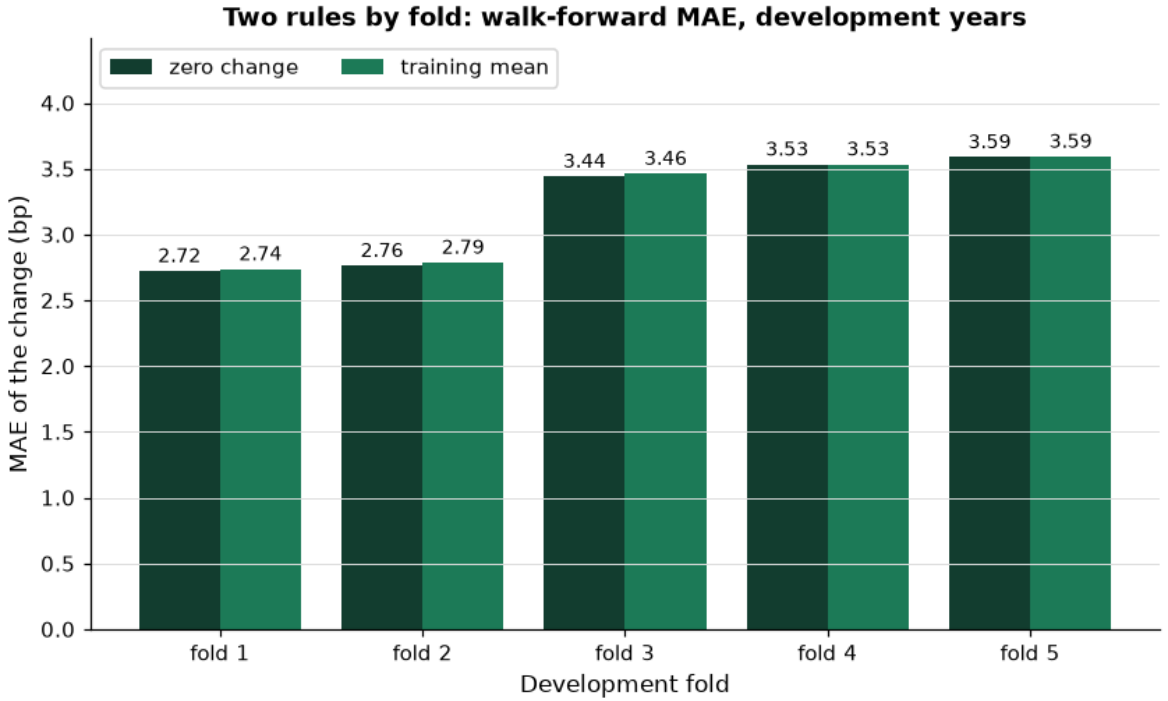
zero change, mean of 5 folds

0 of 5

folds where the training mean is below

Difference +0.013 bp. Fitting one number to the past did no better than fitting nothing: day 11's finding in one line.

The two rules, fold by fold



In every fold the training mean's bar is the taller. The bars rise and fall together: the folds differ in how far the 10 Yr moved.

The words a result may use: assert_history

Course 4: FORECAST_WORDS

Stops on a claim about a result not measured.

Course 5: EXPLANATION_WORDS

Stops on an explanation the numbers do not hold.

Today: HISTORY_WORDS

Stops on a date past the end of the data.

Allowed names

fitted change in bp

walk-forward error

the last value carried forward

Never called

a forecast or a prediction

tomorrow's move or a call

an outlook or a view on rates

59 tests pass: netkit is complete. The lists are a reminder, not a proof.

The case study: which network, and how did it score?

The colleague's question

Of the sequence networks this week built, which would the development years have chosen, and how did that choice score on the held-out years?

1. Refit an LSTM and a GRU of 16 units on day 11's five development folds
2. Choose by a rule written first: the lower mean walk-forward MAE
3. Set the held-out folds from a count; open the held-out years once
4. Measure the seed spread; write the workbook and where it fails

Nothing is chosen after section 14.6. It describes past dates; it is not a view on rates.

The development years: the size of an average day

3.75 bp

mean absolute daily change, 10 Yr

5.03 bp

standard deviation of the change

192 of 2,000

dates with no change at all

Zero change's error is the move

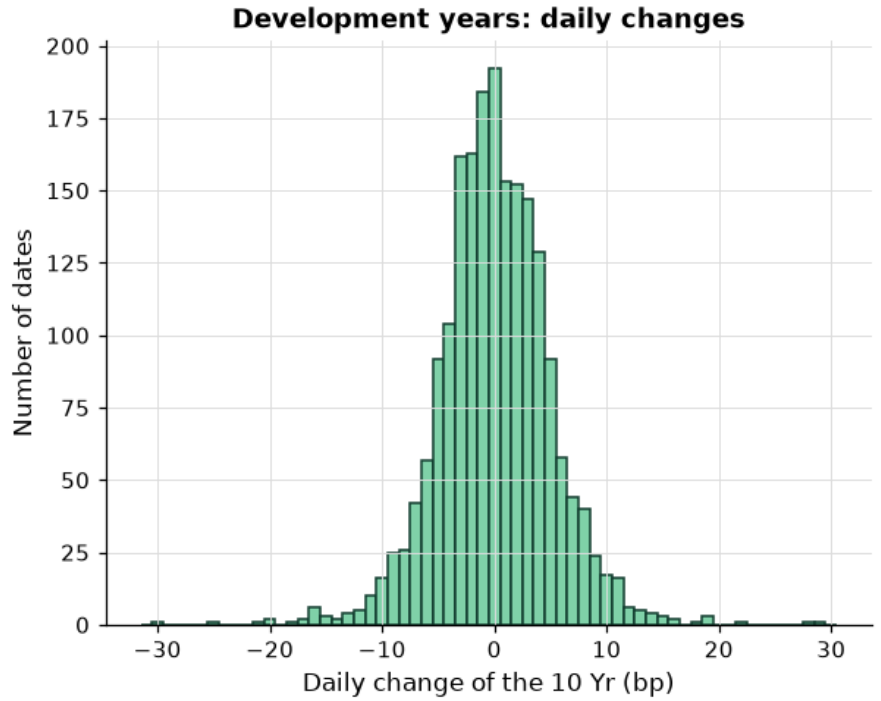
On each date zero change misses by the size of that date's change, so its MAE is the average move.

Two decimals of a percent

Every change is a whole number of bp. On a day with no move, zero change is exactly right.

10 Yr level 0.52 to 4.25 percent. Course 1 notebook 14 and Course 3 notebooks 06 and 07 hold the full picture.

The development changes, drawn





The tallest bar is at 0 bp: 192 of 2,000 dates. The networks are fitted on changes, not on levels.

How the held-out folds work

1. Fold 1

trains on every development date

2. Later folds

add the held-out dates before them

3. Sizes

counted from dates, fixed in advance

4. Together

every held-out date scored once

You control

The number of folds, and through it their size.

What comes out

Three folds, each scored on dates it never saw.

The held-out folds, set from a count



train the first held-out fold

fold 1 fold 2 fold 3

Inputs: day 11's, unchanged

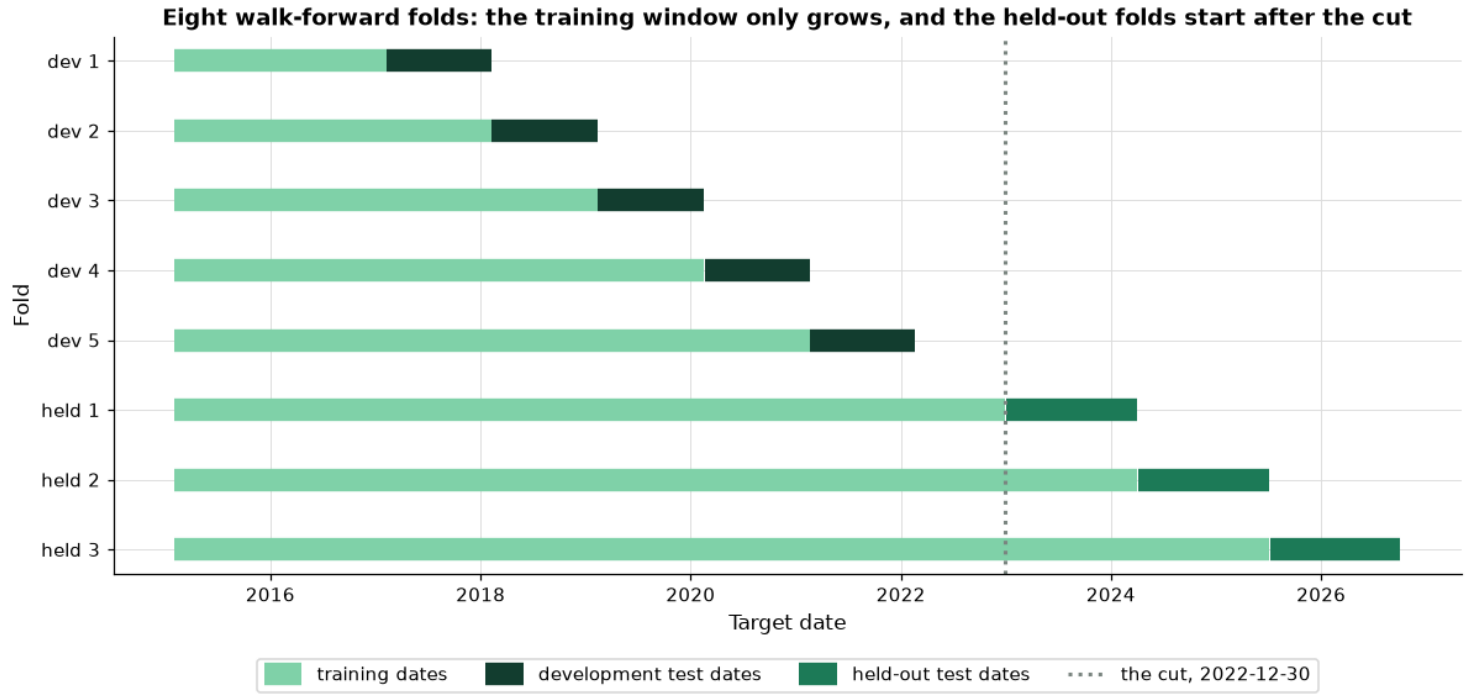
The 20 changes before each date, one channel. Scaler, target, and stopping rows from each fold's training dates only.

3 x 313 = 939

Every held-out date scored once, none dropped. Each fold trains on every date before its first.

Set before a single held-out yield is read.

Eight folds on one timeline



The training window only grows; every training bar ends before its test bar starts. Dates only: no held-out yield is read.

Choosing on the development years

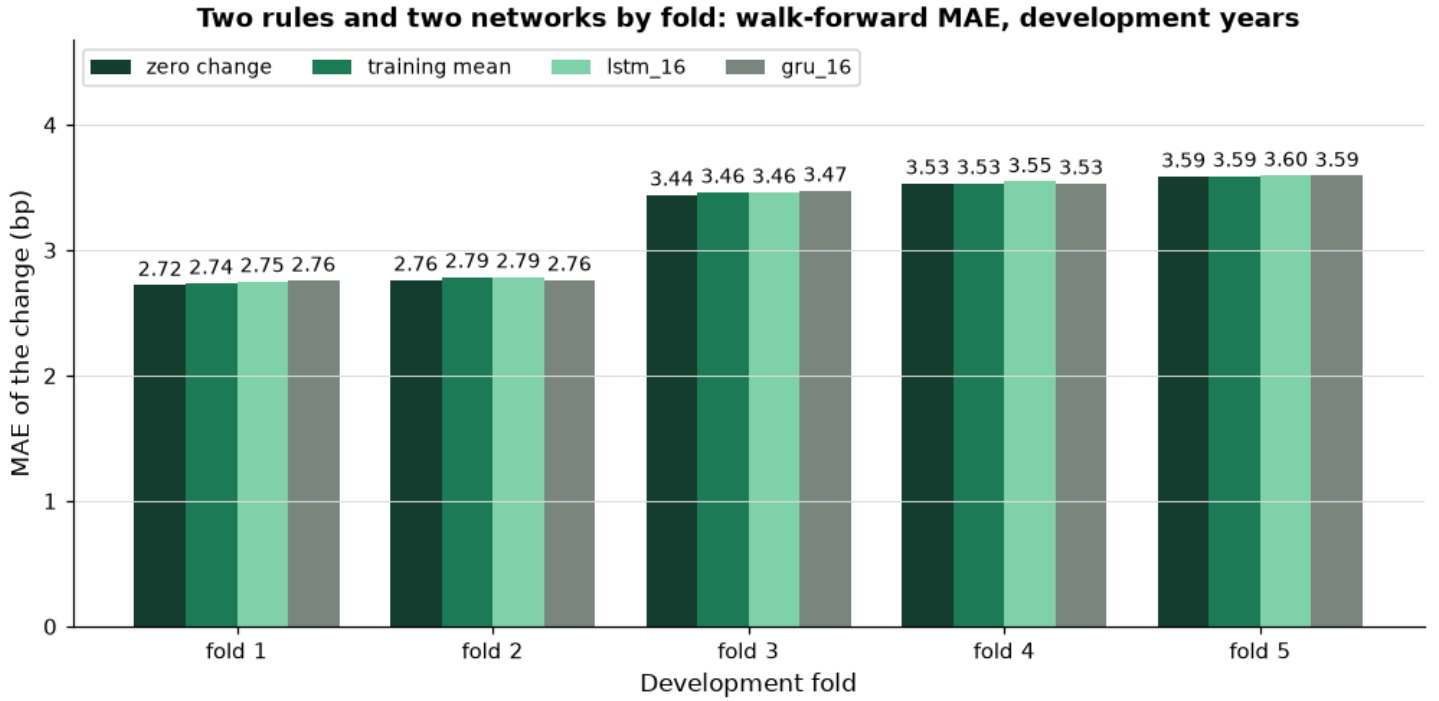
Development folds	Weights	MAE (bp)	Difference	Folds below
Zero change		3.21		
lstm_16	1169	3.23	+0.019	0 of 5
gru_16	929	3.22	+0.015	1 of 5

Decision: gru_16, by 0.005 bp

The rule was written first: the lower mean walk-forward MAE over the development folds. A gap this small a seed could reverse; the choice is the rule's.

Both are above zero change and above the training mean on average.

Two rules and two networks, fold by fold



The four bars of a fold are nearly the same height; zero change's is the shortest on all but one fold.

The held-out years, opened once

Fold	Test dates	gru_16	Zero change	Training mean
fold 1	2023-01-03 to 2024-04-02	5.77	5.75	5.75
fold 2	2024-04-03 to 2025-07-03	4.60	4.57	4.58
fold 3	2025-07-07 to 2026-10-02	3.37	3.37	3.37

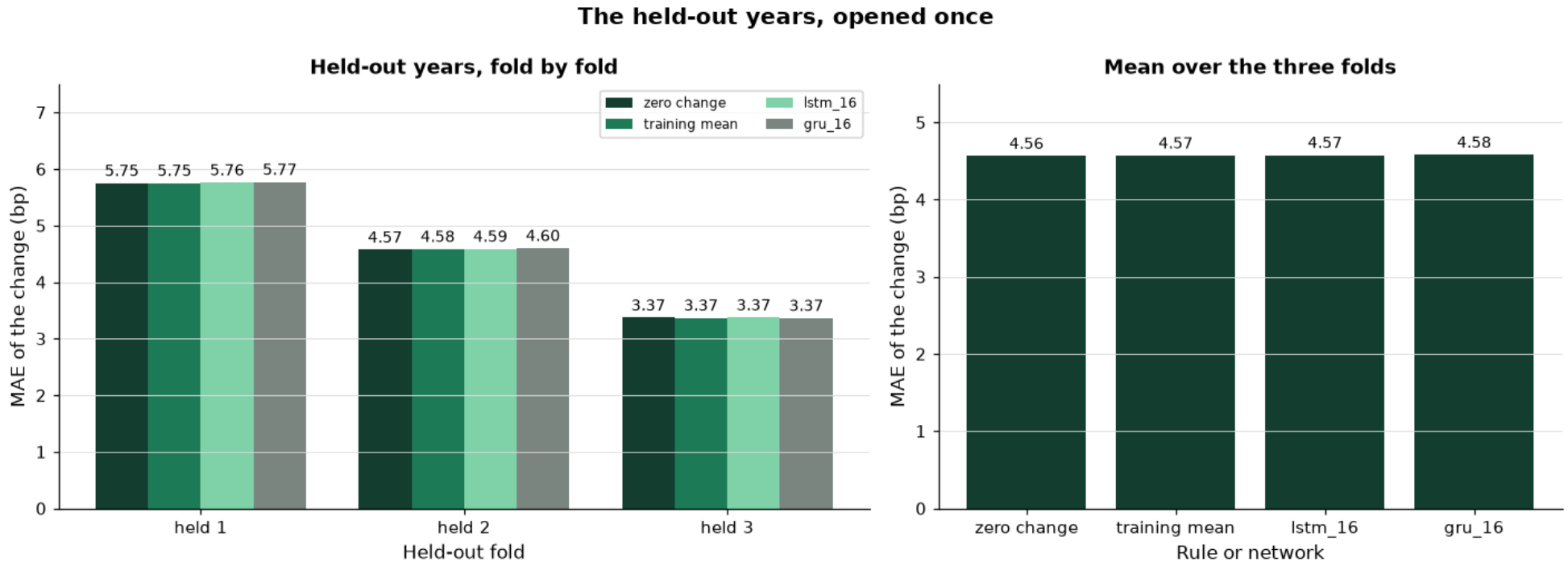
All held-out dates	MAE (bp)	Folds below zero change
Zero change	4.56	
Training mean	4.57	
lstm_16 (not chosen)	4.57	0 of 3
gru_16 (chosen)	4.58	1 of 3

The two networks swapped

Development: GRU lower by 0.005 bp. Held out: LSTM lower by 0.004 bp. The choice stands.

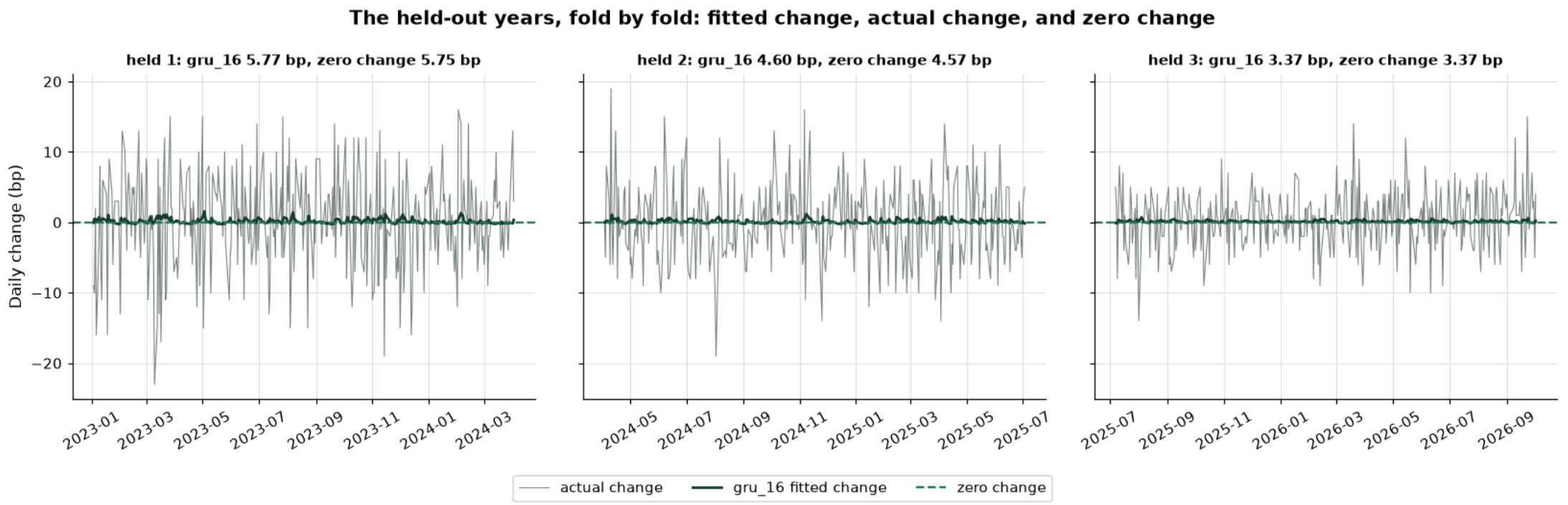
Nothing beat the last value carried forward. The GRU: +0.014 bp against zero change.

The held-out folds, as bars



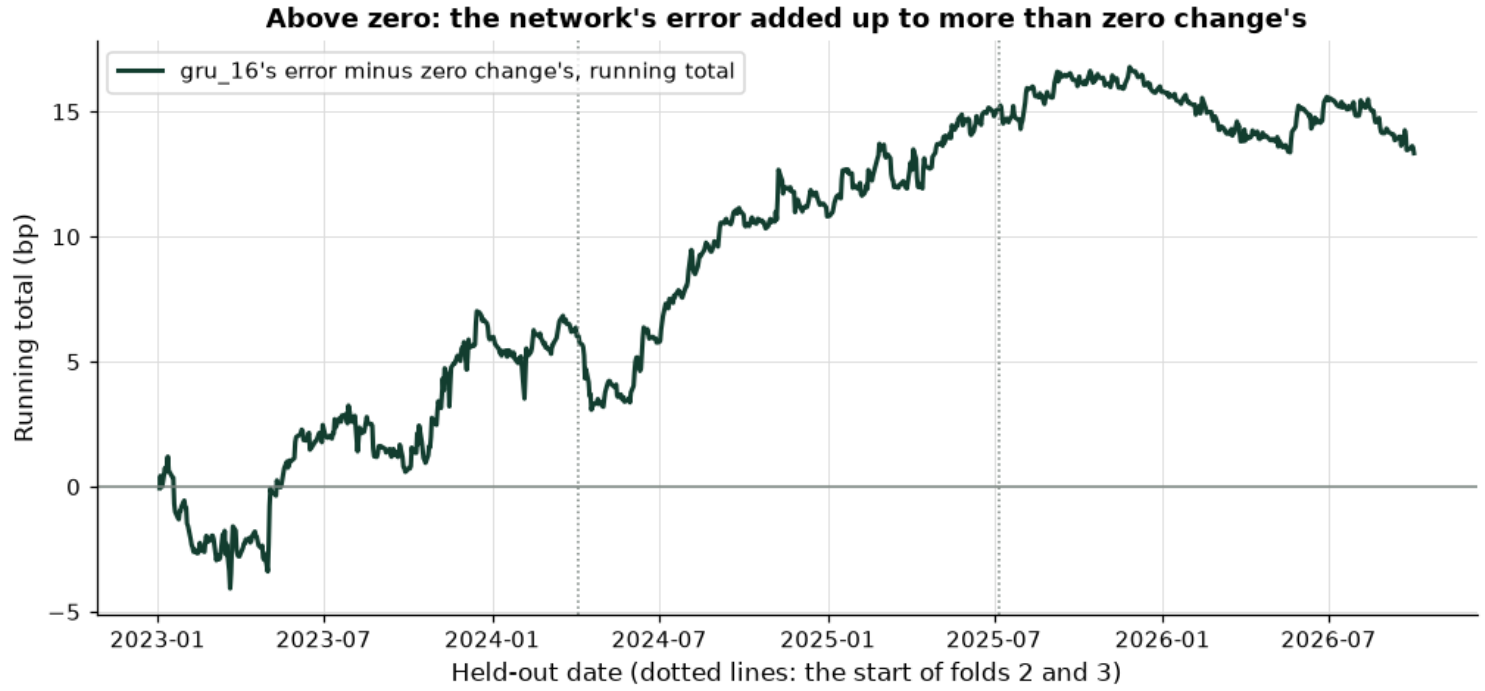
Nothing beat the last value carried forward. The GRU: +0.014 bp against zero change.

The held-out folds, fitted against actual



In every held-out fold the fitted line stays near zero while the actual change swings.

The running total against zero change



Not a straight drift: ahead at first, behind through the middle, and back toward zero at the end.

How the seed spread works

1. A seed

sets the starting weights and batch order

2. Refit

the same network with other seeds

3. Compare

the gap against the spread

4. The rule

a gap inside the spread is luck

You control

The seeds.

What comes out

One score per seed, and their range.

How much the seed alone moves the chosen network

Seed	Held-out MAE (bp)	Minus zero change (bp)	Folds below
42	4.579	0.014	1
43	4.581	0.017	0
44	4.587	0.022	0

0.008 bp

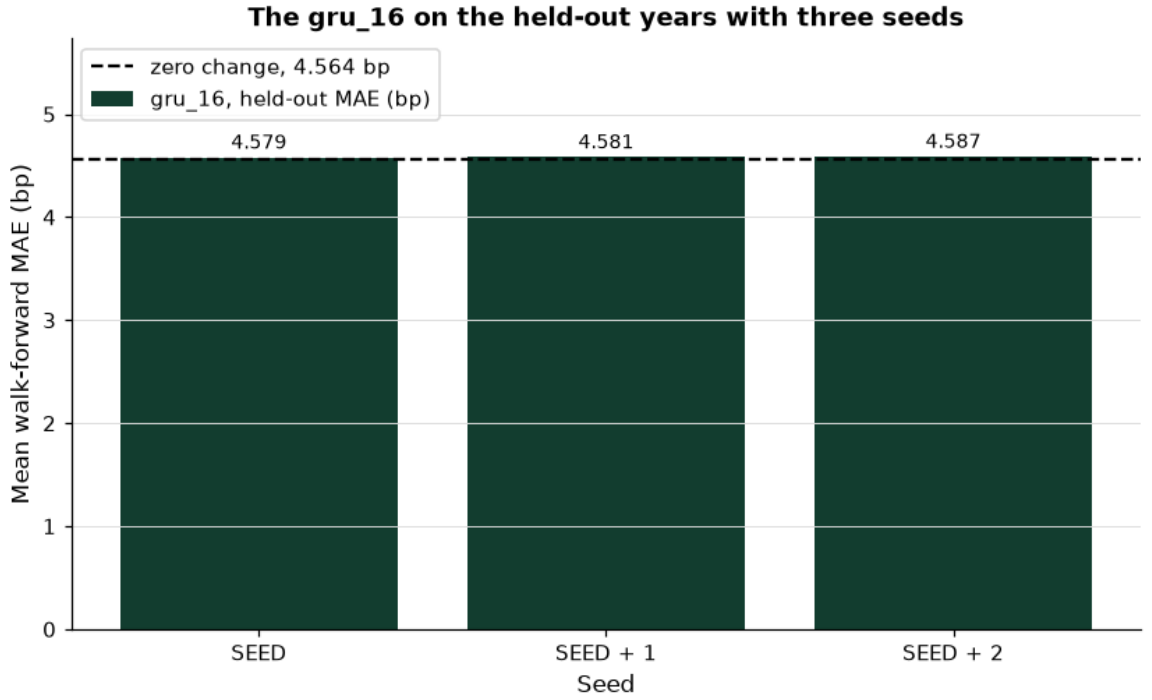
range across three seeds

0.004 bp

held-out gap, GRU against LSTM

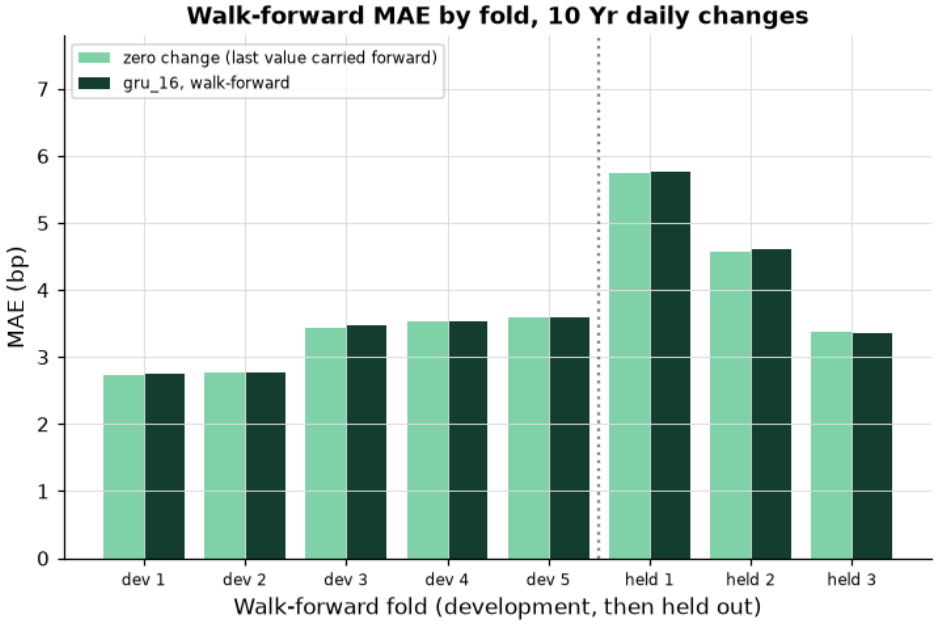
The seed moves more than the gap: the two networks cannot be told apart. Above zero change with 3 of 3 seeds.

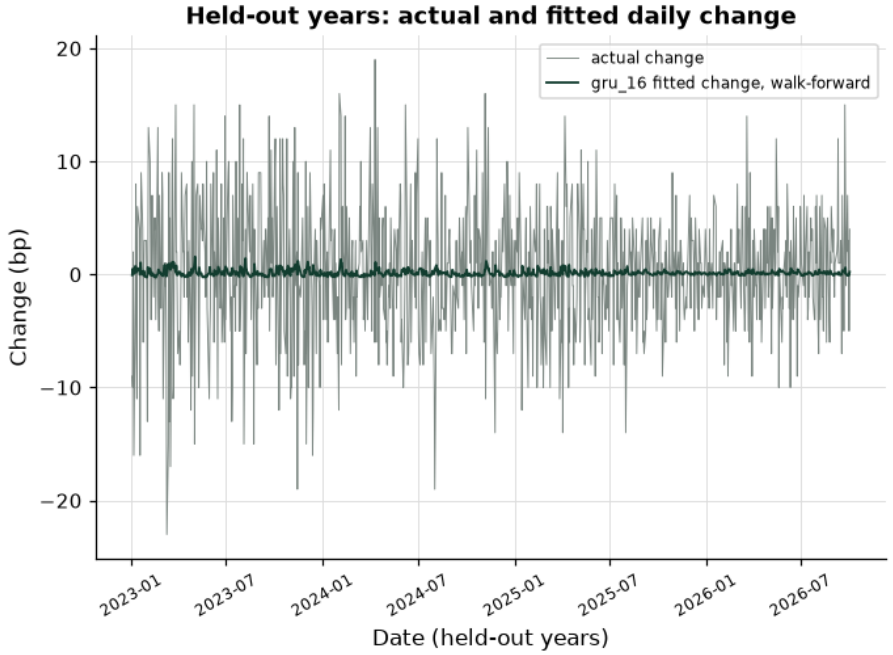
Three seeds against zero change



All three bars end at the dashed line to the eye; the table shows each above it by thousandths. The seed moves the bar by up to 0.008 bp; nothing a seed does brings it down to the line.

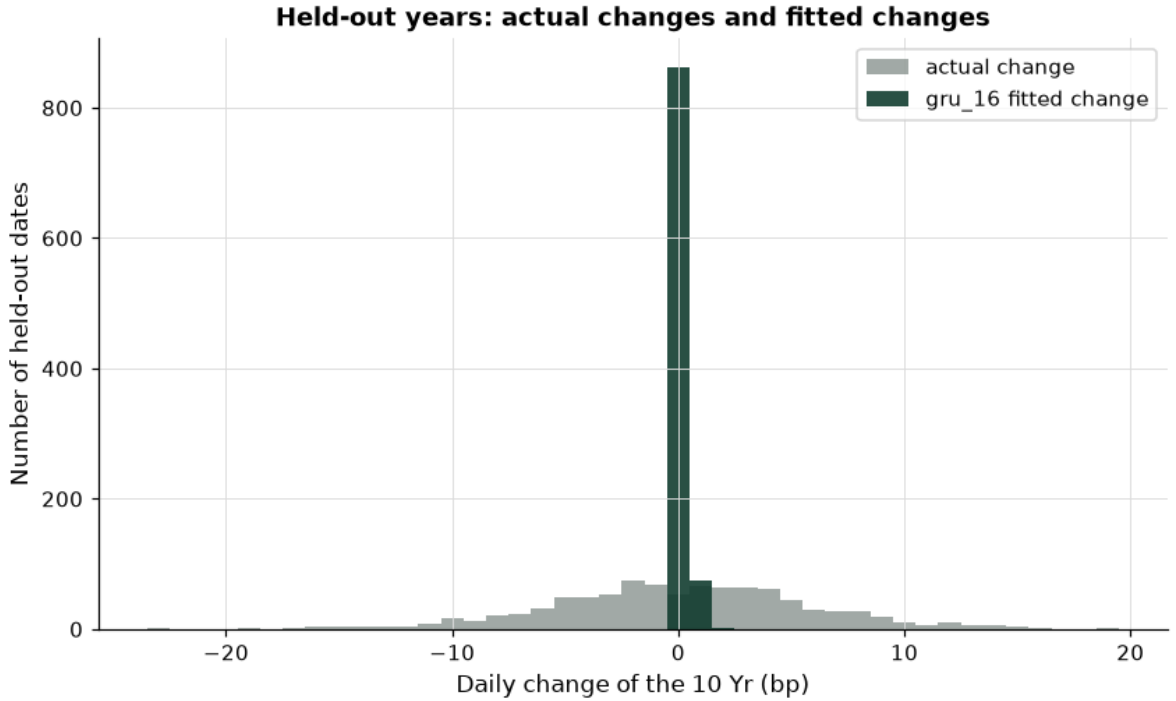
Every fold, and every held-out date





Fitted changes: standard deviation 0.26 bp against 5.83 bp for the actual changes. The fit barely leaves zero.

Fitted against actual, as histograms



The fitted changes are a narrow spike near zero: standard deviation 0.26 bp against 5.83 bp.

What the model shows, and where it fails

	Development	Held out
Mean absolute change (bp)	3.75	4.56
Standard deviation (bp)	5.03	5.83
10 Yr range (percent)	0.52 to 4.25	3.30 to 5.29

What it shows

Chosen in advance, the GRU was further from the change than zero change, in all but one fold and with every seed.

On the dates

Fitted changes barely leave zero; worse than zero on 484 of 939 dates.

On the choice

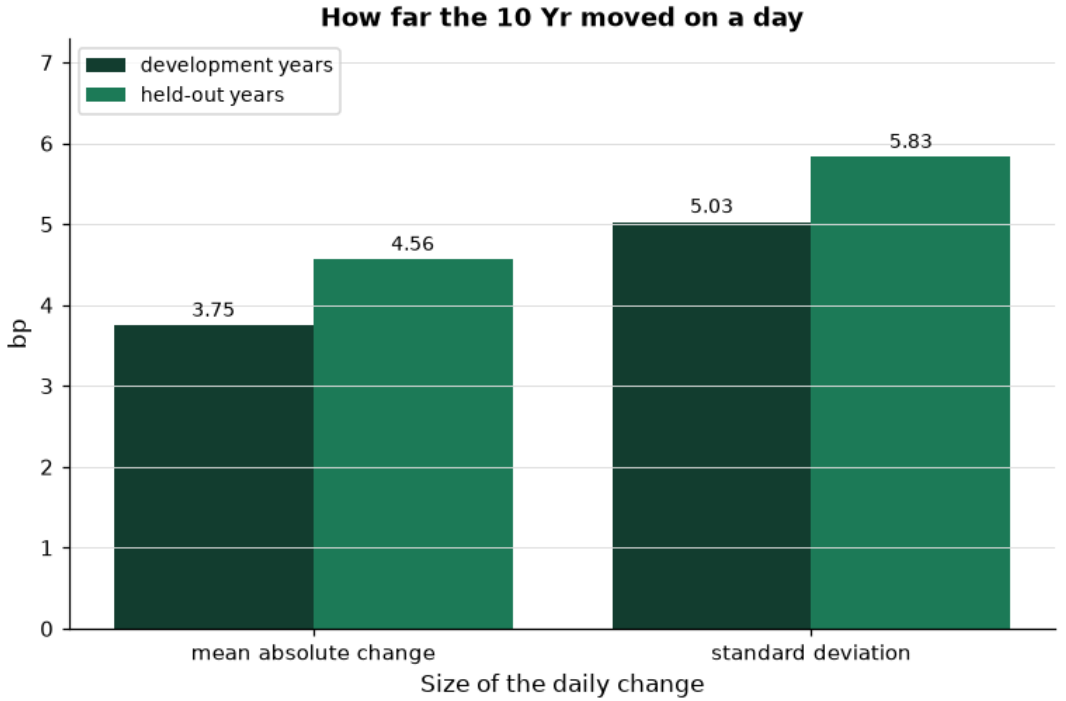
Chosen by 0.005 bp, behind by 0.004 bp held out; the seed moves 0.008 bp.

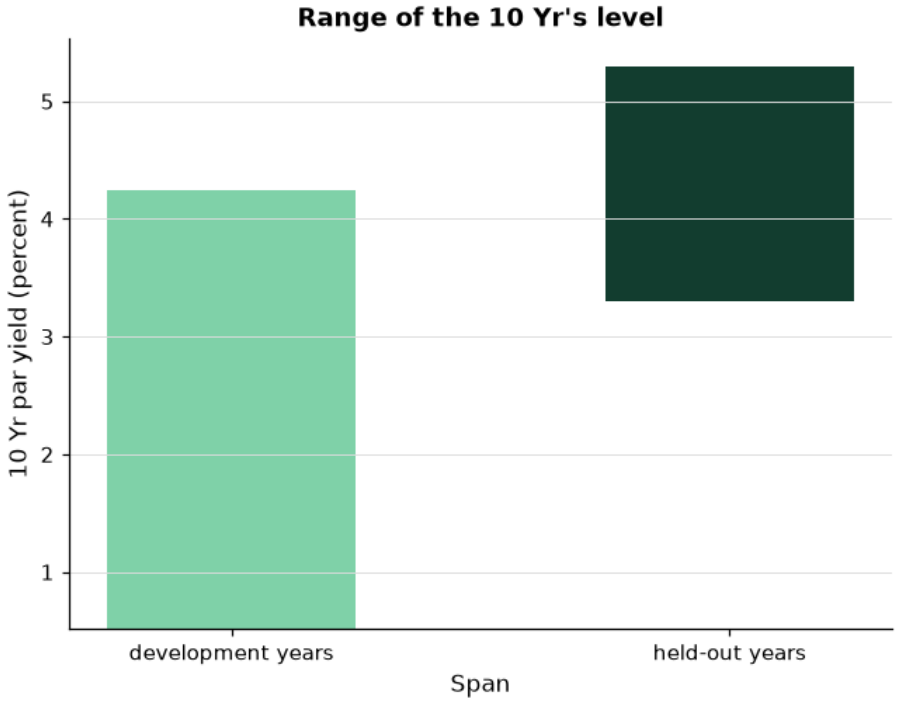
On the span

Fitted on calmer years, scored on busier ones; 20 days of one tenor.

A walk-forward error on these past dates is not evidence about any later date, and not a view on rates.

The two spans, as bars





Both size bars are taller for the held-out years: a network fitted on calmer years was scored on busier ones.

Excel side by side

Excel side by side (tested in Excel)

`=ABS(C2-D2)`	the network's error on a date, E2 filled down
`=ABS(C2)`	zero change's error, F2 filled down
`=AVERAGEIFS(held_out!$E$2:$E$940,held_out!$B$2:$B$940,A2)`	network MAE, fold 1: 5.7700
`=AVERAGEIFS(held_out!$F$2:$F$940,held_out!$B$2:$B$940,A2)`	zero change MAE, fold 1: 5.7508
`=IF(B2<C2,1,0)`	below zero change: 0, 0, 1
`=COUNTIF(D2:D4,1)/COUNT(D2:D4)`	share of folds below: 0.3333

No walk-forward routine and no network training in Excel. Every number above equals Python's within 1e-12.

Save the day: week3-teaching

1. Code

netkit complete

2. Development

four rows

3. Held out

four rows below

4. Tag

week3-teaching

```
model,channels,window,rows,folds,mae_bp,zero_change_mae_bp,folds_below_zero_change
zero_change,1,20,development,5,3.2087,3.2087,0
training_mean,1,20,development,5,3.2216,3.2087,0
lstm_16,1,20,development,5,3.2281,3.2087,0
gru_16,1,20,development,5,3.2234,3.2087,1
zero_change,1,20,held_out,3,4.5644,4.5644,0
training_mean,1,20,held_out,3,4.5688,4.5644,1
lstm_16,1,20,held_out,3,4.5743,4.5644,0
gru_16,1,20,held_out,3,4.5786,4.5644,1
```

git diff week3-teaching~1 week3-teaching shows the four held_out rows, nothing above them changed.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

Docstring

Which rows fit, which rows stop, which rows report: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert what the docstring promises. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the final model is chosen on the held-out years: lstm_16 at 4.57 bp, against gru_16 at 4.58 bp.

Code that runs is not the same as code that is right.

What you can do now

- Compare a model with a baseline by fold
- Check a description with assert_history
- Choose by a rule written before opening
- Open the held-out years once, walk-forward
- Say when the seed outweighs a gap

Week 3, Day 5 The capstone: a network against LightGBM